Safety systems · Austria

Austrian main signals say whether and how fast you may pass (**Halt**, **Frei**, **Frei mit 40**, **Frei mit 60**); distant signals (**Vorsignale**) what the next main signal shows. Other speeds come from speed indicators beside the signal. On the **Semmeringbahn** (1116, 4024) **PZB 90** watches the signals; on the **Mittenwaldbahn** (1020, 1989) it is the older **Indusi I60**, and the German section to the border has German signals (see the Germany manual). Speeds in km/h. Quadrant controls are from TheJAG's ÖBB profiles: the same layout as the German trains.

Safety systems

PZB 90
Punktförmige Zugbeeinflussung

Magnets at **1000 Hz** (distant signal), **500 Hz** (before some main signals; rare in Austria) and **2000 Hz** (main signal). Acknowledge, then keep within the speed curve.

WHERE Semmeringbahn: 1116, 4024

QUADRANT **ENG 2** Wachsam, **ENG 1** Frei, **round aux** Befehl 40; right aux square PZB on / off

Indusi I60
Induktive Zugsicherung (1989)

After **Wachsam** (4 s) be at or below 95 (O) / 75 (M) km/h within **34 s**: one check, no curve. 500 Hz: a spot check at 40. **Frei only releases a penalty brake.**

WHERE Mittenwaldbahn: 1020

QUADRANT Same buttons: ENG 2 Wachsam, ENG 1 Frei, round aux Befehl; right aux Indusi

SiFa
Sicherheitsfahrschaltung

1116 / 4024: after 30 s a lamp, 2.5 s later a buzzer, 2.5 s later the emergency brake. **1020: no lamp**: a buzzer after 30 s, an alarm 75 m later, the brake 75 m after that.

WHERE Every ÖBB train

QUADRANT **Gear lever** = SiFa reset; left aux square SiFa on / off

LZB / ETCS
Continuous systems

LZB on parts of the Westbahn and ETCS on new lines in the real world; neither on these TSW routes.

WHERE –

Main signals (Hauptsignale)

1980 type shown (Semmeringbahn). The 1954 type on the Mittenwaldbahn is one column, top to bottom green, red, yellow, green, with the same combinations.



Halt
Stop.

DRIVE Stop before the signal; 2000 Hz live.



Frei
Proceed at the permitted speed.

DRIVE Drive on.



Frei mit 40
Green over yellow: max 40 km/h.

DRIVE Be at 40 at the signal.



Frei mit 60
Two greens: max 60 km/h.

DRIVE Be at 60 at the signal.



Ersatzsignal
Flashing white: pass the signal at Halt.

DRIVE Hold Befehl over the 2000 Hz; max 40 through the points.

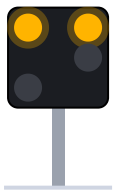


Vorsichtssignal
Flashing red: pass on sight.

DRIVE On sight, max 30 (TSW: 40).

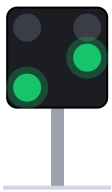
Distant signals (Vorsignale)

On the Mittenwaldbahn every Vorsicht and "frei mit 40 / 60" must be acknowledged, magnet or not.



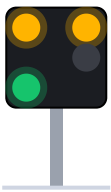
Vorsicht
The main signal shows Halt.

DRIVE Wachsam, brake to stop.



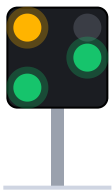
Hauptsignal frei
The main signal shows Frei.

DRIVE Keep going.



Hauptsignal frei mit 40
The main signal shows Frei mit 40.

DRIVE Wachsam, brake to 40.



Hauptsignal frei mit 60
The main signal shows Frei mit 60.

DRIVE Wachsam, brake to 60.

Sources: Dovetail Games / TSG: Mittenwaldbahn manual · ÖBB 30.02 Signalbuch, 30.01 Betriebsvorschrift V3 · de.wikipedia: Eisenbahnsignale in Österreich, PZB, Sicherheitsfahrschaltung · Commons: Diagrams of railway signals in Austria

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Speed signals and boards

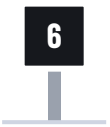
Digits are tens of km/h.



Geschwindigkeits-voranzeiger

With a distant signal: the main signal's speed will be 60.

DRIVE Brake to 60 by the main signal.



Geschwindigkeits-anzeiger

With a main signal: 60 from here; overrules the signal's speed.

DRIVE Be at 60 at the signal.



Ankündigungstafel

Line speed changes ahead (700–1000 m).

DRIVE Brake to the new speed by the Geschwindigkeitstafel.



Geschwindigkeitstafel

Line speed from here; the next tafel can raise it.

DRIVE Lower: by the board. Higher: once the rear has passed.



Ankündigungssignal

Temporary restriction (Langsamfahrstelle) ahead.

DRIVE Brake to the speed by the Anfangssignal.



Anfangssignal

Start of the Langsamfahrstelle.

DRIVE Be at the speed here.



Endsignal

End of the Langsamfahrstelle.

DRIVE Accelerate once the rear has passed.



Abstandstafel

Three, two, one stripes: the distant signal is coming.

DRIVE Watch for the distant signal.

PZB 90 (1116, 4024) and Indusi I60 (1020)

Magnet	PZB 90	Indusi I60
1000 Hz at a distant at Vorsicht	Wachsam within 4 s; curve by train type over 1250 m; Frei after 700 m once the signal has cleared	Wachsam within 4 s; ≤ 95 (O) / 75 (M) km/h within 34 s; the lamp goes out by itself
500 Hz (rare)	Falling curve to 45 / 35 / 25 km/h; Frei not possible	Spot check: ≤ 40 km/h
2000 Hz at Halt	Immediate emergency brake; Befehl 40 held with an order	Immediate emergency brake; Befehl held at ≤ 40 km/h
After a penalty brake	Stop, Frei, release; restrictive mode where a check runs	Brake pipe at ~0 bar, then Frei, release

How to drive

Approaching Vorsicht

1. PZB / Indusi sounds at the distant signal: press **Wachsam** (ENG 2) within 4 s.
2. Brake: PZB 90 by its curve; Indusi to 95 / 75 within 34 s.
3. Stop before the main signal at Halt.

Passing Halt with an order

1. Ersatzsignal or a written order only.
2. Hold **Befehl** (round aux) while the front passes the 2000 Hz magnet, at max 40.
3. Max 40 through the points.

SiFa

1. Reset with the **gear lever** when it warns (1116 / 4024 lamp; 1020 buzzer).
2. Missed: emergency brake; stop, reset, release.

Before departure: All three load with SiFa and PZB / Indusi off: switch both on (aux squares lit). Set the train type: in the cab on the 1116 / 4024; on the 1020 the Indusi mode knob is in the compartment near cab 2.

Sources: Dovetail Games / TSG: Mittenwaldbahn manual · ÖBB 30.02 Signalbuch, 30.01 Betriebsvorschrift V3 · de.wikipedia: Eisenbahnsignale in Österreich, PZB, Sicherheitsfahrschaltung · Commons: Diagrams of railway signals in Austria

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